
Prompt Forge Installation Guide

React + Spring Boot + Python Application Installation Guide

Prerequisites

You'll need to install some software on your computer. Think of these as the "tools"

For Manual Installation

Required Software:

- 1. Node.js (for React frontend)**
 - What it is: JavaScript runtime that lets you run React applications
 - Version needed: 16.0 or higher
- 2. Java Development Kit (JDK) (for Spring Boot backend)**
 - What it is: Needed to run Java applications
 - Version needed: JDK 11 or higher
- 3. Python (for Python services)**
 - What it is: Programming language runtime
 - Version needed: Python 3.8 or higher
- 4. Git (for code management)**
 - What it is: Tool to download and manage code
 - Installation: Download and install with default settings

For Docker Installation

Required Software:

1. Docker Desktop

- Installation: Download and install for your operating system
 - After installation: Start Docker Desktop and wait for it to be "running"
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Manual Installation

Step 1: Download the Application Code

1. Open your Command Prompt (Windows) or Terminal (Mac/Linux)
 - Navigate to where you want to store the application: `cd Desktop` # or any folder you prefer
2. Download the code:
 - `git clone https://github.com/COS301-SE-2025/Prompt-Forge cd [YOUR_PROJECT_FOLDER]`

Step 2: Set Up the React Frontend

1. Navigate to the React application folder: `cd frontend`
2. Install dependencies:
 - `npm install`
3. This may take a few minutes as it downloads required packages.
4. Start the React application: `npm run dev`
5. Your browser should automatically open to <http://localhost:5137>
6. Keep this terminal window open - closing it will stop your frontend.

Step 3: Set Up the Spring Boot Backend

1. Navigate to the Spring Boot application folder: `cd backend/project`
2. Create a file called `.env` in the backend (secrets granted upon request)
3. Start the SpringBoot application:
 - For Windows: `./mvnw spring-boot:run`
 - If mvnw doesn't work, and you have Maven installed: `mvn spring-boot:run`
 - For Mac/Linux: `./mvnw spring-boot:run`
4. Wait for the message: "Started Application in X.XXX seconds"
5. The backend will be available at <http://localhost:8080>
6. Keep this terminal window open - closing it will stop your backend.

Step 4: Set Up the Python Service

1. Navigate to the Python application folder: `cd path/to/your/project/python-service`
adjust path as needed
 2. Create a virtual environment: `python -m venv venv`
 3. Activate the virtual environment:
 - For Windows: `venv\Scripts\activate`
 - For Mac/Linux: `source venv/bin/activate`
 4. Install Python dependencies: `pip install -r requirements.txt`
 5. Start the Python service: `python app.py`
 6. The Python service will be available at <http://localhost:5000>
 7. Keep this terminal window open - closing it will stop your Python service.
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Docker Installation

Step 1: Download the Application Code

1. Download the code:
 - `git clone https://github.com/COS301-SE-2025/Prompt-Forge cd [YOUR_PROJECT_FOLDER]`

Step 2: Verify Docker Installation

1. Make sure Docker is running:
 - a. `docker --version`
 - b. `docker-compose --version`

Step 3: Set Up Environment Configuration

1. Create a file called `.env` in the backend (secrets granted upon request)

Step 4: Build and Start All Services

1. `docker-compose up --build`

Step 5: Access Your Application

Once you see messages indicating all services are ready:

- Frontend (React): `http://localhost:5173`

- Backend (Spring Boot): `http://localhost:8080`
- Python Service: `http://localhost:8000`

Step 6: Stop the Application

1. When you're done, stop all services: Press `Ctrl+C` in the terminal, then run:
2. **`docker-compose down`**

Verification

Check If Everything Is Working

1. **Frontend Check:**
 - Open `http://localhost:3000` in your browser
 - You should see your React application interface
2. **Backend Check:**
 - Open `http://localhost:8080/health` (or your health endpoint)
 - You should see a JSON response indicating the service is up
3. **Python Service Check:**
 - Open `http://localhost:8000/health` (or your health endpoint)
 - You should see a response indicating the service is running
4. **Integration Check:**
 - Use the frontend to perform an action that requires backend data

Troubleshooting

Common Issues and Solutions

- **"Port already in use" Error**

Problem:

Another application is using the required port.

Solution:

- Find and stop the conflicting application, or
- Change the port in your configuration files
- **"Command not found" Error**

Problem:

Required software isn't installed or not in PATH.

Solutions:

- Reinstall the missing software
- Restart your terminal/command prompt
- Check that software is added to your system PATH
- **"Permission denied" Error**

Problem:

Insufficient permissions to run commands.

Solutions:

- Windows: Run Command Prompt as Administrator
- Mac/Linux: Use sudo before commands
- **Docker Issues**

Problem:

Docker containers won't start.

Solutions:

- Make sure Docker Desktop is running
- Check available disk space (Docker needs several GB)
- Restart Docker Desktop
- Run docker system prune to clean up old containers
- **Database Connection Issues**

Problem:

Cannot connect to database.

Solutions:

- Verify database credentials in .env files

- Ensure database service is running
- Check network connectivity

Getting Help

If you encounter issues not covered here:

1. Check the application logs for specific error messages
 2. Search for the error message online
 3. Check our project's README file for additional instructions
 4. Contact the development team with:
 - Your operating system
 - The exact error message
 - What you were trying to do when the error occurred
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Support Information

Technical Requirements:

- Minimum RAM: 4GB (8GB recommended)
- Disk Space: 2GB free space minimum
- Operating System: Windows 10+, macOS 10.15+, or modern Linux distribution
- Internet Connection: Required for initial setup and package downloads

Estimated Setup Time:

- Manual Installation: 30-60 minutes (first time)
- Docker Installation: 15-30 minutes (first time)
- Subsequent starts: 2-5 minutes

For technical support or questions about this installation guide, please contact the development team or refer to the project documentation.